Row of buses - Array (2D) and columns

Row of blocks - Array

Roads connecting - Graph
different places

Branches of tree - Tree

Q.) Create a structure variable student and estimate the size of the structure manually and write a C program to find the size of the structure using size of operator. Now observe both the o/p. Are they same or diff, justify your answer.

* Self referential structure are those structures in which one or more pointers points to the structure of the same type

```
struct self
{
int p;
struct self *pt;
};
```

Q1) manually

int → 4 bytes

char → 1 byte

float → 4 bytes

Program:

```
#include<stdio.h>
struct student{
int id;
char grade;
float gpa;
};
int main(){
struct student s1;
Printf("sizeof student %lu\n", sizeof(s1));
```

Observation:

upon running it, we will see 12 bytes instead of manual 9 bytes

Justification:

* Structure Padding:

In this case, the compiler likely adds 3 padding bytes after the char member so that the float begins on a 4-byte boundary